A few years later, this prediction, that the radio-emitting lobes are powered by jets of gas emerging from a central engine, was spectacularly confirmed using huge new radio interferometers in Britain, Holland, and America—most notably the American VLA *(very large array)* on the plains of St. Augustin in New Mexico (Figure 9.5). The interferometers saw the jets, and the jets had just the predicted properties. They reached from the galaxy's core to the two lobes, and they could even be seen ramming into gas in the lobes and being slowed to a halt.

The VLA uses the same "spots on the bowl" technique as the radio interferometers of the 1940s and 1950s (Figure 9.2), but its bowl is much larger and it uses many more spots (many more linked radio telescopes). It achieves resolutions as good as 1 arc second, about the same as the world's best optical telescopes—a tremendous achievement when one contemplates the crudeness of Jansky's and Reber's original instruments forty years earlier. But the improvements did not stop there. By the early 1980s, pictures of the cores of radio galaxies and quasars, with resolutions 1000 times better than optical telescopes, were being produced by *very long baseline inteiferometers* (VLBIs) composed of radio telescopes on opposite sides of a continent or the world. (The output of each telescope in a VLBI is recorded on magnetic tape, along with time markings from an atomic clock, and the tapes from all the telescopes are then played into a computer where they are "interfered" with each other to make the pictures.)